

svm with hard margin and C value

Asked 11 years, 2 months ago Modified 11 years, 2 months ago Viewed 8k times



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I have to create a a svm classifier with RBF kernel and a hard margin. If I undertood correctly, the kind of margin is controlled by the C-parameter. That is, the larger C parameter, the more hard margin I will be getting. Is there a typical C value that emulates a very robust "hard margin"? Or is there any other way to emulate a robust hardmargin? Of course we could always discuss why do we need to emulate a hard margin. But in this case I need to emulate it!

```
clf = svm.SVC(C = ??,kernel="rbf",gamma =1.5,shrinking=False,cache_size=3000)
```

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edited Sep 10, 2012 at 22:03



[Fred Foo](#)

356k

78

744

836

asked Sep 10, 2012 at 16:06



[andreSmol](#)

1,028

2

18

30

- 2 You could try setting it to some absurdly large value like $C=10e20$; I'm not sure how high you can set C without running into overflows, though. – [Fred Foo](#) Sep 10, 2012 at 22:04

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Usually you don't need that large values, but in general that is the idea. Just try setting it to $1e10$ and then look at the resulting alphas.

If none of the alphas is at the C value, no regularization was applied (i.e. all points are classified correctly in the trainingset = hard margin).

Btw, if your dataset if very noisy and can not be separated with the kernel you are using, you can not achieve a hard margin.

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answered Sep 11, 2012 at 8:37

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[Andreas Mueller](#)



Thanks Andreas! I have a question about getting the alphas, if I am using SVC, there are no methods to get the alphas and the actual support vectors. We have to read the "internal" parameters `support_support_vector_`. Is it correct? Why there are no methods?? – [andreSmol](#) Sep 11, 2012 at 15:21



There are no methods because the alphas are stored in an attribute, `dual_coef_`, see [the docs](#). The `support_` and `support_vector_` are explicitly documented and exposed to the user. The

trailing underscore is a scikit-learn convention that says that these attributes are estimated from the data. – [Andreas Mueller](#) Sep 11, 2012 at 15:23
